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Dear Editor,

I wish to submit the article “Every Algebraic Number is a Critical Value of an Integer Polynomial” for consideration by The American Mathematical Monthly. This work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. A preprint is posted on GitHub.

The main theorem of this paper is in the title, and the construction is explicit. This answers a 2017 question of Schwarz, Vologodsky and Walcher. Petrov arrived at an equivalent construction, without knowing that the question was open. I arrived at Petrov’s theorem by reducing the question to its statement. The main new ingredient compared with Petrov’s argument is arranging integer coefficients, while preserving the prescribed critical value.

I also classify the integer polynomials that can have a given algebraic number as a critical value, and give a lower bound on their degree. A companion website provides an online calculator.

Since this article is short, Professor Dingxing Zhang told me the American Mathematical Monthly would be suitable for it.

Keywords: critical values, integer polynomials, algebraic numbers, Petrov’s theorem, Lean formalization.

I have no conflicts of interest to disclose. Please address all correspondence concerning this manuscript to me at rj@simis.cn. Thank you for your consideration of this manuscript.

Sincerely,

RJ Acuña